

Finite Fredholm Truncation for Long Increasing Subsequences

A proof of fixed-sector holonomicity
and uniform all-orders tail asymptotics

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Abstract

Let $A(N, k)$ count permutations of N letters whose longest increasing subsequence has length exactly k . We prove that, for every fixed integer $r \geq 2$, a globally holonomic array agrees with $A(N, k)$ throughout $N \leq rk$, $k \geq 1$. This establishes the fixed-sector existence conjecture stated by Kauers and Wang in September 2026. The construction uses factorial moments of the number of shifted Young-diagram rows beyond a threshold. A rectangle obstruction makes inclusion–exclusion terminate after $r - 1$ terms in the sector. The discrete Bessel kernel expresses each fixed moment as a finite-dimensional proper-hypergeometric sum, so holonomic closure applies. We determine the exact first failure of each truncation.

We also derive a single-sum formula for the first moment, prove uniform all-orders asymptotics when k is a positive linear fraction of N , and compute five correction terms for OEIS A269021. The error from replacing the tail by its first shifted-row moment is either zero or smaller than every algebraic order. Consequences include rational-ray P-recursiveness, conditional overshoot estimates, and a mean-subsequence-multiplicity limit. Exact enumeration and symbolic checks are supplied separately.

Status and scope. This is an unrefereed conventional mathematical proof, not a Lean- or Rocq-verified development. The discrete Bessel-kernel and holonomic-summation theorems are established external inputs. The specific previously guessed recurrences in the one-third sector are not certified here; the theorem proved is their underlying existence statement. Priority beyond the sources checked is not asserted.

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1 Problem, provenance, and main results

1.1 The arrays and the question

For $N \geq 0$ and $k \geq 1$, set

$$A(N, k) = \#\{\pi \in S_N : \text{LIS}(\pi) = k\}, \quad (1.1)$$

$$T(N, k) = \#\{\pi \in S_N : \text{LIS}(\pi) \geq k\}. \quad (1.2)$$

Here an increasing subsequence need not occupy adjacent positions. These are the exact-length array [A047874](#) and the upper-tail array [A214152](#), respectively [8, 9]. We use $A(0, k) = T(0, k) = 0$ for $k \geq 1$, and both arrays vanish for $k > N$. They satisfy

$$A(N, k) = T(N, k) - T(N, k + 1). \quad (1.3)$$

The diagonal $T(2n, n)$, with its separately defined empty-permutation value $a_0 = 1$, is [A269021](#); $A(2n, n)$, with the same convention at zero, is [A267433](#) [10, 11].

The distinction between a separate recurrence for each fixed k and a single bivariate recurrence system is essential. A bivariate holonomic array has uniform polynomial-coefficient recurrences in the index directions. In particular, there are nonzero operators of the forms

$$\sum_{i=0}^d p_i(N, k) E_N^i, \quad \sum_{i=0}^e q_i(N, k) E_k^i, \quad p_i, q_i \in \mathbb{Q}[N, k], \quad (1.4)$$

where $E_N F(N, k) = F(N + 1, k)$ and $E_k F(N, k) = F(N, k + 1)$. Orders and degrees do not grow when N and k vary. We use “holonomic” in the discrete holonomic-systems sense, for which finite sums, products, and definite summation have their usual closure properties. These properties imply the D-finiteness required here.

Following the regional convention in [1], an array is *D-finite in a region* if it agrees there with a globally D-finite array. This does not say that a recurrence stencil may leave the region while its entries are still taken from the original array.

Kauers and Wang proved the half-sector result and the previously conjectured fourth-order recurrence for [A269021](#) (Conjecture 17 of [2]). Their September 2026 paper asks for D-finiteness in every fixed sector $N \leq rk$, $r \geq 2$ [1, Sections 1–2]. We prove this existence statement, rather than claiming their already established recurrence as new. The main device is a truncation by the number of shifted rows, not by the number of ordinary increasing-subsequence witnesses.

1.2 The main theorem

For a partition $\lambda = (\lambda_1, \lambda_2, \dots)$, let f^λ be the number of standard Young tableaux of shape λ . Extend its parts by zeros and define

$$Y_k(\lambda) = \#\{i \geq 1 : \lambda_i - i \geq k - 1\}, \quad C_j(N, k) = \sum_{\lambda \vdash N} (f^\lambda)^2 \binom{Y_k(\lambda)}{j}. \quad (1.5)$$

The sum defining Y_k is finite because $k \geq 1$.

Theorem 1.1 (Finite-moment holonomic extensions). *For every fixed $j \geq 1$, $C_j(N, k)$ is a holonomic bivariate array. For fixed $R \geq 1$, define*

$$T_R(N, k) = \sum_{j=1}^R (-1)^{j+1} C_j(N, k), \quad (1.6)$$

$$A_R(N, k) = T_R(N, k) - T_R(N, k+1). \quad (1.7)$$

Both arrays are globally holonomic and

$$T_R(N, k) = T(N, k), \quad A_R(N, k) = A(N, k) \quad \text{if } N < (R+1)(k+R). \quad (1.8)$$

Consequently, for every fixed $r \geq 2$, the arrays A and T are D -finite throughout $N \leq rk$, $k \geq 1$: take $R = r-1$.

The word “globally” refers to the whole nonnegative N , positive k quadrant, not just to the agreement sector. A zero column may be added at $k = 0$ without changing holonicity. The values of T_R and A_R outside their agreement region are auxiliary values; in particular, positivity outside the region is not required.

The gain is not merely one additional sector. The same construction works for every fixed r , and its summation dimension depends only on r , not on N . At the same time, it does *not* prove global holonicity of the original array A : the required R is unbounded as k/N tends to zero.

Theorem 1.2 (Uniform tail expansion). *Put $m = N - k$, $\rho = m/k$, and*

$$B(N, k) = \frac{(N!)^2}{(k!)^2(N-k)!}. \quad (1.9)$$

For every compact interval $I \subset (0, \infty)$ and integer $M \geq 0$, uniformly over integer pairs with $k \rightarrow \infty$ and $\rho \in I$,

$$\frac{T(N, k)}{B(N, k)} = e^{-2\rho} \left(\sum_{d=0}^M \frac{c_d(\rho)}{k^d} + O_{I,M}(k^{-M-1}) \right). \quad (1.10)$$

The coefficients are explicitly computable polynomials, $c_0 = 1$, $\deg c_d \leq 2d$, and

$$c_1(\rho) = 2\rho - \rho^2, \quad (1.11)$$

$$c_2(\rho) = \frac{\rho(3\rho^3 - 20\rho^2 + 42\rho + 2)}{6}, \quad (1.12)$$

$$c_3(\rho) = -\frac{\rho(\rho^5 - 14\rho^4 + 73\rho^3 - 158\rho^2 + 26\rho + 30)}{6}. \quad (1.13)$$

An all-orders coefficient rule is given in Section 7.

These are Poincaré asymptotic expansions with a remainder for every fixed truncation order, not a claim of convergence of the infinite series. Section 6 identifies a distinct, beyond-all-algebraic-orders contribution from diagrams with a second long shifted row. We do not claim a full expansion of that exponentially small contribution.

1.3 Relationship to earlier work and the repository

The leading A269021 asymptotic $16^n(n-1)!/(\pi e^2)$ is already recorded in the OEIS and attributed to Václav Kotesovec in 2016 [10]. The same leading equivalent for A267433 has a public proof by David Moews from

2023 [7]. The contribution here is the uniform all-orders development, together with the fixed-sector theorem and its sharp truncation boundary, not the discovery of that leading constant.

The ProveIt repository was inspected at revision 6bf7f30d0352f7596e70928b3d4f304914075907 [12]. A targeted identifier search did not surface an A269021 report. That is a bounded search observation, not an exhaustive assertion about every artifact or an external priority search. This article uses no unverified mathematical assertion from unrelated repository projects. It is designed as a self-contained research package suitable for later integration.

The new deductions below rely explicitly on three standard inputs: the Robinson–Schensted enumeration, the discrete Bessel correlation formula, and holonomic summation. The remaining reductions, cutoff analysis, finite-sum formulas, and asymptotic error estimates are proved in the article. The package includes no claim of minimal recurrence order.

[write] Added 3 October 2026, batch 85. *Provenance.* This report is a single manuscript, printed in full: manuscript 08 of batch 85 of ProveIt’s incoming-reports intake, delivered as the archive LIS_fixed_sector_research_package.zip (wrapper directory LIS_fixed_sector_research/) in commit 9d6968c8a, and placed in commit ddf8df5d5 in the research-report collection, among the OEIS-sequence asymptotics reports, as the report a047874-long-increasing-subsequences, named after the array whose sector question it answers. Its pin is the revision 6bf7f30d0 quoted above. The title page and the PDF author field read “ChatGPT”: the package names ChatGPT as the tool that wrote it and names no human author. Every label now carries the prefix lis::; no statement, proof or number of the manuscript was changed.

What Kauers and Wang state. The arXiv abstract of [1] does not mention the sector question, so on 3 October 2026 the intake read the HTML text of arXiv:2609.02220v1 (licensed CC BY 4.0). Its Section 1 defines “D-finite inside a region” as the existence of a D-finite array coinciding with the original one there (the convention used above), proves two bivariate recurrences valid for $n/2 \leq k$ (their Theorem 1, for their $a_{n,k}$, which is $A(N, k)$ here) and derives Conjecture 17 of [2] from them. At the end of their Section 2, after displaying guessed recurrences of higher order and degree for $n/3 < k < n/2$, they write: “We suspect that for every $r = 2, 3, \dots$, the sequence $a_{n,k}$ is D-finite in the range $k \geq n/r$ ”; they add that this would not give D-finiteness for all $k \geq 0$, and hope for a uniform approach to all such slices. So the statement is an informal conjecture, not a numbered one; “the fixed-sector existence conjecture” is this manuscript’s name for it. Its content is exactly the sector assertion of Theorem 1.1 ($k \geq n/r$ is $N \leq rk$). For $r = 2$ the D-finiteness is already Kauers and Wang’s Theorem 1, with explicit recurrences; there Theorem 1.1 is only a second route to existence. The new content is every $r \geq 3$ and the affine cutoff (1.8). Kauers and Wang’s guessed $r = 3$ operators and all minimal orders remain uncertified (Section 4.3).

Review status. Nothing here is refereed or formalized. The holonomic-closure step of Section 3.4 is the step most in need of independent review; see the note there. The intake checked the finite identities independently: a separate hook-length program reproduced (5.4) for all $N \leq 14$, $1 \leq k \leq N + 1$, the truncation identity (1.8) and the first failure (2.5) for $R = 1, 2, 3$ in that range, and the diagonal values of the table in Section 9.2 for $n \leq 6$.

Repository. The statement above that a targeted search “did not surface an A269021 report” is true at the pin and was confirmed at placement: no other file of ProveIt treats A047874, A214152, A269021 or A267433, and no Lean or Rocq development treats Young tableaux or longest increasing subsequences. Other conjectures of [2] are treated in the collection reports enumerative-combinatorics/a181280-binary-matrix-formula (Conjecture 20), enumerative-combinatorics/a195806-hexagonal-lattice (Conjecture 11) and Part IV of a215561-fixed-composition-excursions (Conjecture 15); Conjecture 17 is Kauers and Wang’s, not this report’s. The batch-85 sibling a181199-shifted-rectangles treats fixed-height shifted rectangles. The rectangle formula (2.6) is the classical hook-length count, printed also as t2:eq:tableaux in the Transseries volume Combinatorial_Transseries_Inverses.

Notation. No symbol was renamed. Some letters carry two meanings:

Symbol	Meaning here	Other meaning, or tempting false reading
$A(N, k), T(N, k)$	exact-length and tail arrays (A047874, A214152)	Kauers and Wang write $a_{n,k}$ and $b_{n,k}$
a_n, b_n (Section 9)	$a_n = T(2n, n)$ (A269021), $b_n = A(2n, n)$ (A267433)	not Kauers and Wang's arrays: their $b_{2n,n}$ is a_n here
R	truncation rank	Kauers and Wang's name for a region
j	moment order of C_j	summation index in (5.6)–(5.7) and Section 7
m	summation index in Lemma 5.1 and Theorem 5.2	the deficit $N - k$ in Theorem 1.2 and Section 7
q	$N/(k+2)^2$ in Proposition 6.1	the summation variables q_a of Theorem 3.1; the denominator in Corollary 4.1
C_1	first shifted-row moment	not the mean witness count (Section 5.2); not the constants C_I
L_ℓ, L_n	logarithmic coefficients (7.4)	L_n is the leading term of a_n in Section 9.3

2 Shifted rows and exact inclusion–exclusion

2.1 From permutations to tableaux

The Robinson–Schensted correspondence gives a bijection between permutations and pairs of standard Young tableaux of the same shape, with the longest increasing subsequence equal to the first row length [6]. Hence

$$T(N, k) = \sum_{\substack{\lambda \vdash N \\ \lambda_1 \geq k}} (f^\lambda)^2, \quad \sum_{\lambda \vdash N} (f^\lambda)^2 = N!. \quad (2.1)$$

We use the hook-length formula to compute f^λ in the verifier; the structural proof only needs (2.1) and the standard rectangle dimension formula when describing the first discrepancy.

The shifted row coordinates $\lambda_i - i$ are strictly decreasing. Therefore $Y_k \geq 1$ is precisely $\lambda_1 \geq k$. More generally:

Lemma 2.1 (Rectangle obstruction). *For $j, k \geq 1$,*

$$Y_k(\lambda) \geq j \iff \lambda_j \geq k + j - 1. \quad (2.2)$$

Thus $Y_k \geq j$ forces $|\lambda| \geq j(k + j - 1)$. At equality, the only possible partition is the rectangle $(k + j - 1)^j$.

Proof. Strict decrease of the shifted coordinates means that at least j coordinates meet the threshold exactly when the j th one does. The first j rows then each have at least $k + j - 1$ boxes. Equality in the total size forces these rows to have exactly that length and leaves no further boxes. Conversely, that rectangle meets the threshold. $\square$

This is the source of the correction $j(j - 1)$ in all cutoff formulas. Counting ordinary rows of length at least k would give the wrong threshold and the wrong moment process.

2.2 The residual identity and its sign

For integers $y \geq 1$ and $R \geq 1$, Pascal's identity gives

$$\sum_{j=1}^R (-1)^{j+1} \binom{y}{j} = 1 - (-1)^R \binom{y-1}{R}. \quad (2.3)$$

The identity remains useful when $R \geq y$, using zero for an out-of-range binomial coefficient. For $y = 0$, the sum is zero. Weighting (2.3) by $(f^\lambda)^2$ proves the following exact formula.

Proposition 2.2 (Bonferroni residual). *For all $N \geq 0$, $k, R \geq 1$,*

$$T_R(N, k) - T(N, k) = (-1)^{R+1} \sum_{\substack{\lambda \vdash N \\ Y_k(\lambda) \geq 1}} (f^\lambda)^2 \binom{Y_k(\lambda) - 1}{R}. \quad (2.4)$$

In particular, odd truncations are upper bounds and even truncations are lower bounds for T . The residual vanishes if $N < (R + 1)(k + R)$.

Proof. Equation (2.1) identifies the contribution of the constant 1 on the right side of (2.3) with T . A nonzero residual requires $Y_k \geq R + 1$, and Lemma 2.1 excludes this below the stated size. Every weight in the residual sum is nonnegative, proving its sign. $\square$

The sign statement belongs to the tail truncations. A difference of two Bonferroni bounds need not give a bound of the same sign for the exact-length array away from the special first-discrepancy point.

Proposition 2.3 (The first failure is exact and nonzero). *Let $N_0 = (R + 1)(k + R)$ and $\mu = (k + R)^{R+1}$. Then*

$$T_R(N_0, k) - T(N_0, k) = A_R(N_0, k) - A(N_0, k) = (-1)^{R+1} (f^\mu)^2. \quad (2.5)$$

For a rectangle b^a ,

$$f^{b^a} = (ab)! \frac{\prod_{i=0}^{a-1} i!}{\prod_{i=0}^{a-1} (b + i)!}. \quad (2.6)$$

Thus (1.8) is the sharp universal cutoff for this particular truncation.

Proof. At N_0 , the rectangle in Lemma 2.1 is the only shape with $Y_k \geq R + 1$. It has $Y_k = R + 1$, so the residual binomial coefficient is one. At threshold $k + 1$ the first possible omitted rectangle is larger by $R + 1$, so $T_R(N_0, k + 1) = T(N_0, k + 1)$. Taking differences proves the assertion for A_R . Formula (2.6) is the hook-length formula specialized to a rectangle. $\square$

R	Exact tail expression	Region of exactness
1	C_1	$N < 2k + 2$
2	$C_1 - C_2$	$N < 3k + 6$
3	$C_1 - C_2 + C_3$	$N < 4k + 12$
4	$C_1 - C_2 + C_3 - C_4$	$N < 5k + 20$

In the first line, the error at $N = 2k + 2$ is the square of the $(k + 1)$ st Catalan number, since $f^{(k+1, k+1)}$ equals that number. The affine cutoff is stronger than the homogeneous sector required by the conjecture.

3 The Bessel kernel and a fixed-dimensional factorial sum

3.1 The established determinantal input

For $t > 0$, poissonized Plancherel measure is

$$\mathbb{P}_t(\lambda) = e^{-t} t^{|\lambda|} \frac{(f^\lambda)^2}{(|\lambda|!)^2}. \quad (3.1)$$

It is a probability measure by (2.1). Let $D(\lambda) = \{\lambda_i - i : i \geq 1\} \subset \mathbb{Z}$. The correlation theorem of Borodin, Okounkov, and Olshanski [3, Theorem 2 and Proposition 2.9] states that

$$\mathbb{P}_t(\{x_1, \dots, x_j\} \subset D(\lambda)) = \det[K_t(x_a, x_b)]_{a,b=1}^j, \quad (3.2)$$

for distinct x_a , where

$$K_t(x, y) = \sum_{\ell=1}^{\infty} J_{x+\ell}(2\sqrt{t}) J_{y+\ell}(2\sqrt{t}). \quad (3.3)$$

Only nonnegative x, y are used here. The Bessel series needed below is

$$J_d(2\sqrt{t}) = t^{d/2} \sum_{s \geq 0} \frac{(-1)^s t^s}{s!(d+s)!}, \quad d \geq 0. \quad (3.4)$$

The normalization in (3.1) contains $(N!)^2$, not $N!$. This factor must be restored after coefficient extraction.

Summing correlations over increasing j -tuples in the threshold set $\{k-1, k, \dots\}$ gives its factorial moment:

$$e^{-t} \sum_{N \geq 0} \frac{C_j(N, k)}{(N!)^2} t^N = \sum_{k-1 \leq x_1 < \dots < x_j} \det[K_t(x_a, x_b)]. \quad (3.5)$$

There is no $1/j!$ here: the x -tuples are strictly ordered and hence already represent unordered subsets once each.

3.2 Cauchy–Binet removes the kernel summation

Write $x_a = k - 1 + u_a$ and $\ell_b = 1 + v_b$. Applying Cauchy–Binet to (3.3) yields

$$e^{-t} \sum_{N \geq 0} \frac{C_j(N, k)}{(N!)^2} t^N = \sum_{\substack{0 \leq u_1 < \dots < u_j \\ 0 \leq v_1 < \dots < v_j}} \left(\det[J_{k+u_a+v_b}(2\sqrt{t})]_{a,b=1}^j \right)^2. \quad (3.6)$$

This also explains the Fredholm terminology: these are the elementary symmetric, or Fredholm-expansion, coefficients of the threshold kernel. We truncate the expansion after a fixed number of these coefficients; we do *not* assert that the infinite kernel has finite operator rank.

All operations in (3.6) are coefficientwise finite. Indeed, factoring the powers of $t^{1/2}$ from rows and columns shows that a squared minor has lowest possible degree

$$b = jk + \sum_a u_a + \sum_b v_b. \quad (3.7)$$

Since $u_a, v_a \geq a - 1$, this is at least $j(k + j - 1)$. For a fixed degree N , only finitely many u, v have $b \leq N$. Although an individual Bessel factor may involve a half-integral power, each squared minor has integral powers. Equivalently, work first with $z = \sqrt{t}$ and retain its even powers.

For analytic justification, finite Cauchy–Binet is applied before taking larger index cutoffs. The Bessel series converges absolutely on compact sets and its orders have factorial decay. Thus the finite cutoffs converge locally uniformly. Alternatively, every coefficient stabilizes once the cutoffs exceed its finite degree budget, which gives the same formal identity. No unproved interchange of an asymptotic expansion with an infinite determinant is being used.

3.3 The explicit multisum

For permutations $\sigma, \tau \in S_j$ and nonnegative integer vectors $s = (s_1, \dots, s_j)$ and $q = (q_1, \dots, q_j)$, put

$$h = N - jk - \sum_a u_a - \sum_b v_b - \sum_a (s_a + q_a). \quad (3.8)$$

Theorem 3.1 (Proper-hypergeometric formula). *For fixed $j \geq 1$, the following finite sum holds for all $N \geq 0$, $k \geq 1$:*

$$\begin{aligned} \frac{C_j(N, k)}{(N!)^2} &= \sum_{\sigma, \tau \in S_j} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \sum_{\substack{0 \leq u_1 < \dots < u_j, 0 \leq v_1 < \dots < v_j \\ s_a, q_a \geq 0, h \geq 0}} \frac{(-1)^{\sum_a (s_a + q_a)}}{h!} \\ &\quad \cdot \prod_{a=1}^j \frac{1}{s_a! q_a! (k + u_a + v_{\sigma(a)} + s_a)! (k + u_a + v_{\tau(a)} + q_a)!}. \end{aligned} \quad (3.9)$$

For each pair (σ, τ) there are $4j$ summation variables, independently of N and k .

Proof. Expand the two determinants in (3.6), and then use (3.4) in each of their $2j$ factors. The total initial power is (3.7), because $\sum_a v_{\sigma(a)} = \sum_a v_{\tau(a)} = \sum_b v_b$. Multiplication by e^t inverts the e^{-t} in (3.6). Its coefficient supplying the remaining degree is $1/h!$. This gives (3.9). Every summation index is bounded once $h \geq 0$ and (3.8) are imposed, so the calculation is an identity of finite coefficient sums. $\square$

3.4 Why holonomic summation applies, including the boundaries

We use the standard proper-hypergeometric summation theorem of Wilf–Zeilberger [4, Section 2, Theorem 1 and Corollary A], together with definite-summation closure for holonomic systems [5, Theorem 2 and Section 4]. In the form needed here:

A fixed-dimensional finite sum of proper-hypergeometric terms, with factorials of integer-affine forms and integer-linear summation boundaries, is holonomic in its remaining integer parameters. Finite sums and products of the resulting holonomic arrays remain holonomic.

A proper term has factorial products or quotients, polynomial factors, and constant-to-integer powers; mere rationality of shift quotients is not the only hypothesis being invoked.

To apply this statement concretely, remove strict inequalities by the gap coordinates

$$u_a = a - 1 + \sum_{b=1}^a \alpha_b, \quad v_a = a - 1 + \sum_{b=1}^a \beta_b, \quad \alpha_b, \beta_b \geq 0. \quad (3.10)$$

Then

$$h = N - jk - j(j-1) - \sum_{b=1}^j (j-b+1)(\alpha_b + \beta_b) - \sum_a (s_a + q_a). \quad (3.11)$$

For fixed j , every factorial argument in (3.9) is an integer-affine expression in

$$N, k, \alpha_1, \dots, \alpha_j, \beta_1, \dots, \beta_j, s_1, \dots, s_j, q_1, \dots, q_j.$$

In particular, jk is linear because j is fixed.

One can regard the inner sum as a sum over the cube $\{0, \dots, N\}^{4j}$, setting $1/h! = 0$ for negative integer h . No nonzero term is lost, since every coefficient in the budget (3.11) is positive. This reciprocal-factorial zero convention is the usual terminating hypergeometric convention. The reciprocal-gamma recurrence extends across its integer zeros. Restriction to the nonnegative cube gives ordinary holonomic summation boundaries; alternatively, its indicator has a rational generating function. Boundary terms produced by shifting N or k are included by definite-summation closure and can be homogenized. We do not simply discard such boundary terms.

Thus the closure theorem applies simultaneously with N, k left as parameters. Multiplying by $(N!)^2$ preserves holonomicity, as do the finitely many σ, τ terms. This proves the first assertion of Theorem 1.1. It is stronger than proving P-recursiveness separately after specializing k .

[write] Added 3 October 2026, batch 85: needs independent review. This subsection carries the whole existence claim, and the intake did not verify it; it should be reviewed independently before the sector theorem is relied on (the package's own audit also says that independent review "remains appropriate"). A reviewer should check at least: that the summand of (3.9), in the gap coordinates (3.10), is holonomic as a function of all $4j + 2$ integer variables, N and k included, under the stated conventions (the reciprocal factorial $1/h!$ vanishing for $h < 0$, and the restriction to $\alpha_b, \beta_b, s_a, q_a \geq 0$, which is imposed by a cube rather than by reciprocal factorials for α_b, β_b); that the closure theorem cited from [4, 5] applies with the N -dependent support cut out by $h \geq 0$; and that the boundary terms of creative telescoping, said above to be "included by definite-summation closure and [able to] be homogenized", are in fact accounted for. The executable checks test the identity of Theorem 3.1 (through the 140 Bessel-minor comparisons) and the finite truncation identities on recorded ranges; they test no holonomicity. Sections 5–9 do not depend on this step: the single sums, the uniform expansion and the A269021 and A267433 corrections use Theorem 5.2, Proposition 2.2 and Proposition 6.1, not holonomic closure. The rational-ray Corollary 4.1 of Section 4 does depend on it.

Completion of Theorem 1.1. Finite linear combinations and shifts give holonomicity of T_R and A_R . Proposition 2.2 gives agreement of T_R with T below $(R + 1)(k + R)$. The same inequality also implies agreement at $k + 1$, so (1.3) gives agreement for A_R . For $R = r - 1$ and $N \leq rk$,

$$N \leq rk < r(k + r - 1) = (R + 1)(k + R),$$

which proves the sector assertion. □

4 Consequences for recurrence theory

4.1 Rational rays and periodic rounding

Corollary 4.1 (Every positive rational ray). *Fix integers $a \geq b \geq 1$ and offsets a_0, b_0 . The sequences*

$$A(an + a_0, bn + b_0), \quad T(an + a_0, bn + b_0)$$

are eventually P-recursive, and any finite completion at small indices is P-recursive. The same conclusion holds for $A(N, \lfloor pN/q \rfloor + c)$ and $T(N, \lfloor pN/q \rfloor + c)$, where $0 < p \leq q$ and c are fixed integers.

Proof. Choose fixed R with $a < (R + 1)b$. The ray lies in the exactness region for all sufficiently large n . Substitution in (3.9) leaves a fixed-dimensional proper-hypergeometric sum with integer-affine factorial arguments in n and the summation indices. Hence its sequence is P-recursive by the same summation theorem. A finite modification preserves P-recursiveness. For the rounded ray, split N into its q residue classes. Each subsequence is a linear ray, and interlacing finitely many P-recursive sequences preserves P-recursiveness: their generating functions combine as $\sum_{s=0}^{q-1} z^s F_s(z^q)$. □

No assertion about irrational slopes follows from this argument. Nor does it impose an effective small order on a recurrence along a ray. The dimension is fixed but can still be computationally expensive.

4.2 An abstract transfer principle

The argument isolates a reusable fact. Suppose a nonnegative integer statistic Y has weighted factorial moments $C_j(N, k)$ that are holonomic for every fixed j , and suppose $Y \geq j$ is impossible below a known size $w_j(k)$. Then the weighted event count for $Y \geq 1$ is represented by a holonomic array on $N < w_{R+1}(k)$, for each fixed R . The proof is exactly (2.3).

The nontrivial application here is that the determinantal correlations make the moments fixed-dimensional even while k grows, and that the Young-diagram obstruction provides the explicit function $w_j(k) = j(k + j - 1)$. A growing-size determinant by itself would not justify bivariate holonomicity.

4.3 What has and has not been certified

The construction proves the existence of the required uniform recurrences and supplies a factorial-sum input from which creative telescoping can, in principle, compute them. It does not show that any particular previously guessed operator belongs to the annihilator. To certify such an operator, one must produce telescoping certificates or compare a sufficiently complete holonomic system with controlled initial and singular data. Checking many values alone is not that proof.

A useful next computation would use $C_1 - C_2$ rather than the entire original array. This has a fixed eight-variable inner sum for C_2 , with four signed determinant-expansion terms. Cancellation, determinant structure, or symmetry may reduce this substantially before elimination. The present existence proof does not depend on successfully performing that large elimination.

5 A single sum for the first moment

5.1 Collapsing the trace

At $j = 1$, (3.6) simplifies to

$$e^{-t} \sum_{N \geq 0} \frac{C_1(N, k)}{(N!)^2} t^N = \sum_{d \geq k} (d - k + 1) J_d(2\sqrt{t})^2. \quad (5.1)$$

Indeed, exactly $d - k + 1$ pairs $(u, v) \in \mathbb{Z}_{\geq 0}^2$ satisfy $u + v = d - k$.

Lemma 5.1 (Two elementary coefficient identities). *For integers $m \geq d \geq 0$,*

$$[t^m] J_d(2\sqrt{t})^2 = \frac{(-1)^{m-d}}{(m!)^2} \binom{2m}{m-d}. \quad (5.2)$$

For $m \geq k \geq 1$,

$$\sum_{d=k}^m (d - k + 1) (-1)^{m-d} \binom{2m}{m-d} = (-1)^{m-k} \binom{2m-2}{m-k}. \quad (5.3)$$

Proof. Squaring (3.4) gives a sum over $s + q = m - d$. After multiplication by $(m!)^2$, its unsigned summand is $\binom{m}{s} \binom{m}{m-d-s}$. Vandermonde's convolution gives (5.2). For (5.3), set $L = m - k$ and $a = m - d$. The left side is

$$\sum_{a=0}^L (L - a + 1) (-1)^a \binom{2m}{a} = [x^L] \frac{(1-x)^{2m}}{(1-x)^2},$$

which is the right side. □

Theorem 5.2 (First-moment single sum). *For every $N \geq 0$ and $k \geq 1$,*

$$C_1(N, k) = (N!)^2 \sum_{m=k}^N \frac{(-1)^{m-k} \binom{2m-2}{m-k}}{(m!)^2 (N-m)!}. \quad (5.4)$$

It equals $T(N, k)$ whenever $N \leq 2k + 1$. In the same region,

$$A(N, k) = (N!)^2 \sum_{m=k}^N \frac{(-1)^{m-k} \binom{2m-1}{m-k}}{(m!)^2 (N-m)!}. \quad (5.5)$$

Proof. Insert Lemma 5.1 in (5.1) and multiply by e^t . Coefficient extraction gives (5.4). Proposition 2.2 gives the agreement region. Subtract the $k + 1$ sum from the k sum; their opposite signs and Pascal's identity give (5.5). $\square$

In particular, for $n \geq 1$,

$$T(2n, n) = ((2n)!)^2 \sum_{j=0}^n \frac{(-1)^j \binom{2n+2j-2}{j}}{((n+j)!)^2 (n-j)!}, \quad (5.6)$$

$$A(2n, n) = ((2n)!)^2 \sum_{j=0}^n \frac{(-1)^j \binom{2n+2j-1}{j}}{((n+j)!)^2 (n-j)!}. \quad (5.7)$$

These formulas are suitable for exact rational arithmetic, with an integer result. They provide an independent computational route to the two OEIS diagonals without enumerating all partitions of $2n$.

5.2 A normalization with a combinatorial meaning

The quantity $B(N, k)$ in (1.9) can also be written

$$B(N, k) = \binom{N}{k}^2 (N-k)!. \quad (5.8)$$

It counts pairs consisting of a permutation and a distinguished increasing subsequence of length k : choose its positions and its values, place those values in increasing order, and permute the remaining values arbitrarily.

Thus T/B is the reciprocal of the mean number of such witnesses conditional on their existence. The auxiliary moment C_1 , however, counts shifted rows, not witnesses. Confusing these two first moments would remove the factor $e^{-2\rho}$ from the asymptotic answer.

6 Why higher shifted-row moments are beyond all orders

The exact first-moment formula remains useful beyond its exact tail region. The next estimate controls the discrepancy without a contour integral or a saddle-point argument.

Proposition 6.1 (Explicit first-moment error bound). *If $N < 2k + 2$, then $C_1(N, k) = T(N, k)$. If $N \geq 2k + 2$ and $q = N/(k+2)^2 < 1$, then*

$$0 \leq \frac{C_1(N, k) - T(N, k)}{B(N, k)} \leq \frac{N}{k} (1-q)^{-2} \frac{(k!)^2 (N-k)!}{((k+1)!)^4 (N-2k-2)!}. \quad (6.1)$$

For $\rho = (N - k)/k$ in any fixed compact $I \subset (0, \infty)$, there are constants $C_I, c_I > 0$ such that, for sufficiently large N ,

$$0 \leq \frac{C_1(N, k) - T(N, k)}{B(N, k)} \leq C_I \exp(-c_I N \log N). \quad (6.2)$$

Proof. The $R = 1$ residual is a sum of $(f^\lambda)^2(Y_k - 1)$ over $Y_k \geq 1$. It is nonzero only if $Y_k \geq 2$, equivalently $\lambda_2 \geq k + 1$. Also $Y_k \leq N/k$, by Lemma 2.1. Therefore

$$0 \leq C_1 - T \leq \frac{N}{k} \sum_{\substack{\lambda \vdash N \\ \lambda_2 \geq k+1}} (f^\lambda)^2. \quad (6.3)$$

Write $\lambda = (\ell_1, \ell_2, \nu)$ and $\nu = N - \ell_1 - \ell_2$. Choose the labels for the first two rows and for the remaining diagram, and ignore all column restrictions involving the first two rows. The rows have forced increasing order after their labels are chosen, so

$$f^{(\ell_1, \ell_2, \nu)} \leq \frac{N!}{\ell_1! \ell_2! \nu!} f^\nu.$$

Using $\sum_{\nu \vdash \nu} (f^\nu)^2 = \nu!$ and discarding restrictions on ν gives

$$\sum_{\lambda_2 \geq k+1} (f^\lambda)^2 \leq (N!)^2 \sum_{\substack{\ell_1, \ell_2 \geq k+1 \\ \ell_1 + \ell_2 \leq N}} \frac{1}{(\ell_1!)^2 (\ell_2!)^2 (N - \ell_1 - \ell_2)!}. \quad (6.4)$$

We even dropped $\ell_1 \geq \ell_2$, which can only enlarge the sum. Increasing either ℓ_i by one multiplies its summand by at most $N/(k+2)^2 = q$. The double sum is consequently bounded by its $\ell_1 = \ell_2 = k+1$ term times $(1-q)^{-2}$. Substituting this bound in (6.3) and dividing by B proves (6.1).

For the uniform conclusion, write $k \geq \alpha_0 N$ with $\alpha_0 > 0$ depending only on I . Then $q = O_I(N^{-1})$, and

$$\frac{(k!)^2 (N - k)!}{((k+1)!)^4 (N - 2k - 2)!} \leq \frac{N^{k+2}}{(k!)^2 (k+1)^4} \leq \frac{N^2}{(k+1)^4} \left(\frac{e^2 N}{k^2} \right)^k. \quad (6.5)$$

Here $k! \geq (k/e)^k$ follows by comparing $\sum_{i=1}^k \log i$ with the integral of $\log x$. The logarithm of the right side of (6.5) is at most $-\alpha_0 N \log N + O_I(N)$. This implies (6.2); in the other region the error is exactly zero. $\square$

The estimate separates two logically different issues. The holonomic sector theorem requires several moments to obtain an *exact* answer. The all-orders algebraic asymptotic expansion requires only the first moment, because every omitted contribution is negligible compared with every power of k^{-1} . This explains why the same asymptotic coefficient family works on both sides of $N = 2k$.

7 Uniform all-orders expansion and its coefficient calculus

7.1 The normalized finite product

Use $m = N - k$ for the deficit and replace the summation index in (5.4) by $k + j$. Division by B gives

$$\frac{C_1(N, k)}{B(N, k)} = \sum_{j=0}^m (-1)^j \binom{2k + 2j - 2}{j} \frac{(m)^{\underline{j}}}{((k+1)^{\bar{j}})^2}. \quad (7.1)$$

Writing $m = \rho k$, this is

$$\frac{C_1}{B} = \sum_{j=0}^m \frac{(-2\rho)^j}{j!} Q_{k,j}(\rho), \quad (7.2)$$

where

$$Q_{k,j}(\rho) = \frac{\prod_{a=0}^{j-1} \left(1 + \frac{2j-2-a}{2k}\right) \left(1 - \frac{a}{\rho k}\right)}{\prod_{a=1}^j \left(1 + \frac{a}{k}\right)^2}. \quad (7.3)$$

Empty products are one. For each fixed j , $Q_{k,j} \rightarrow 1$. The alternating exponential series in (7.2) therefore suggests $e^{-2\rho}$, but uniform summation and all-order error control must still be justified.

7.2 An explicit algorithm for every coefficient

Define the following polynomial in j , with coefficients in $\mathbb{Q}[\rho^{-1}]$:

$$L_\ell(j; \rho) = \frac{(-1)^{\ell+1}}{\ell} \left\{ \sum_{a=0}^{j-1} \left(\frac{2j-2-a}{2} \right)^\ell + \sum_{a=0}^{j-1} \left(-\frac{a}{\rho} \right)^\ell - 2 \sum_{a=1}^j a^\ell \right\}. \quad (7.4)$$

Faulhaber's formulas interpret each finite power sum polynomially. Set

$$P_0(j; \rho) = 1, \quad P_d(j; \rho) = \frac{1}{d} \sum_{\ell=1}^d \ell L_\ell(j; \rho) P_{d-\ell}(j; \rho). \quad (7.5)$$

This is the coefficient recurrence for exponentiating $\sum_{\ell \geq 1} L_\ell u^\ell$. Finally, define

$$c_d(\rho) = e^{2\rho} \sum_{j=0}^{\infty} \frac{(-2\rho)^j}{j!} P_d(j; \rho). \quad (7.6)$$

The last expression is elementary, not an unevaluated infinite-sum obstruction. If $P_d(j; \rho) = \sum_b p_{d,b}(\rho) j^b$, then

$$c_d(\rho) = \sum_b p_{d,b}(\rho) \mathcal{T}_b(-2\rho), \quad \mathcal{T}_b(x) = \sum_{a=0}^b \left\{ \begin{matrix} b \\ a \end{matrix} \right\} x^a. \quad (7.7)$$

Here the braces denote Stirling numbers of the second kind. The identity follows by expanding j^b in falling factorials and summing $\sum_j x^j (j)_a / j! = x^a e^x$. No positive-probability interpretation is imposed on the negative argument -2ρ .

For example,

$$L_1(j; \rho) = -\frac{j^2 + 7j}{4} - \frac{j(j-1)}{2\rho}, \quad (7.8)$$

and $P_1 = L_1$. Substitution in (7.7) gives $c_1 = 2\rho - \rho^2$. Equations (7.4)–(7.7) produce all the subsequent coefficients by exact rational arithmetic.

Proposition 7.1 (Polynomial structure). *For every $d \geq 1$, $c_d \in \mathbb{Q}[\rho]$, $c_d(0) = 0$, and*

$$\deg c_d = 2d, \quad [\rho^{2d}] c_d = \frac{(-1)^d}{d!}. \quad (7.9)$$

The continuous value at $\rho = 0$ is a polynomial fact; it does not by itself establish uniformity of the asymptotic expansion there.

Proof. Since $\deg_j L_\ell \leq \ell + 1$, (7.5) gives $\deg_j P_d \leq 2d$. The only negative powers of ρ in (7.3) come from $\prod_{a=0}^{j-1} (1 - au/\rho)$, where $u = 1/k$. The coefficient of ρ^{-b} for $b \geq 1$ vanishes at each integer $j = 0, 1, \dots, b$: there are fewer than b nonzero factors a to select. Thus, as a polynomial in j , that coefficient has a falling-factorial expansion involving only $(j)^a$ with $a \geq b + 1$.

In (7.6), $\rho^{-b}(j)^a$ becomes $\rho^{-b}(-2\rho)^a$, which is polynomial and divisible by ρ . For $b = 0$ and $d \geq 1$, the coefficient vanishes at $j = 0$ because the empty product is identically one, and the same divisibility follows. The degree bound is $2d$.

To obtain degree $2d$, the only term of degree $2d$ in j with no negative power of ρ is $(-j^2/4)^d/d!$ from $L_1^d/d!$. Every L_ℓ with $\ell > 1$ has degree strictly less than 2ℓ . Replacing j^{2d} by its Touchard polynomial contributes the leading term $(-2\rho)^{2d}$. The claimed leading coefficient follows. $\square$

7.3 Uniform remainder estimates

Proof of Theorem 1.2. Fix $0 < \rho_0 \leq \rho \leq \rho_1 < \infty$. First, the absolute value of the j th summand of (7.1) is bounded by

$$\binom{2k+2j-2}{j} \frac{(m)^{\underline{j}}}{((k+1)^{\bar{j}})^2} \leq \frac{[2\rho_1(1+\rho_1)]^j}{j!}, \quad 0 \leq j \leq m. \quad (7.10)$$

Indeed, $j \leq m \leq \rho_1 k$, the top of the binomial is at most $2(1+\rho_1)k$, $(m)^{\underline{j}} \leq (\rho_1 k)^j$, and $(k+1)^{\bar{j}} \geq k^j$. This is a summable bound independent of k .

For a quantitative expansion, split the sum at $J_k = \lfloor k^{1/4} \rfloor$. For $j \leq J_k$, regard (7.3) as a function $Q_j(u; \rho)$ of u on $0 \leq u \leq 1/k$. All logarithmic factors stay away from zero uniformly. Its logarithm has Taylor coefficients (7.4); its ℓ th derivative is bounded by $C_{I,\ell}(1+j)^{\ell+1}$. Also $|\log Q_j(u; \rho)| \leq C_I j^2/k$, so Q_j itself is bounded uniformly on this interval.

Repeated differentiation of an exponential expresses its derivatives as finite sums of products of these logarithmic derivatives. Consequently its $(M+1)$ st derivative is bounded by $C_{I,M}(1+j)^{2M+2}$. Taylor's theorem gives

$$Q_{k,j}(\rho) = \sum_{d=0}^M \frac{P_d(j; \rho)}{k^d} + O_{I,M} \left(\frac{(1+j)^{2M+2}}{k^{M+1}} \right), \quad j \leq J_k. \quad (7.11)$$

After multiplication by $(2\rho)^j/j!$, the remainders sum to $O_{I,M}(k^{-M-1})$ because factorial denominators dominate every fixed power of j .

For $j > J_k$, (7.10) bounds the original summands. The polynomial approximants have the same type of bound with an extra fixed power of j . For any fixed C, D, L , the tail $\sum_{j > k^{1/4}} C^j (1+j)^D/j!$ is $o(k^{-L})$. This follows, for example, from $j! \geq (j/e)^j$ followed by a geometric tail bound once j is sufficiently large. Thus one can extend each coefficient sum to infinity with an error smaller than every algebraic order.

Equations (7.2), (7.11), and (7.6) now give (1.10) with C_1 in place of T . Proposition 6.1 shows that replacing C_1 by T does not change any coefficient or the stated remainder. $\square$

8 Exact-length counts, overshoots, and subsequence multiplicity

The uniform tail expansion also gives a local exact-length expansion. For fixed N and k , replacing k by $k+1$ changes the parameters to

$$\rho' = \frac{\rho k - 1}{k + 1}, \quad \frac{B(N, k + 1)}{B(N, k)} = \frac{\rho k}{(k + 1)^2}. \quad (8.1)$$

Both ratios admit expansions to arbitrary order in $1/k$ on the same compact ρ intervals. Subtraction via (1.3) yields:

Corollary 8.1 (Exact-length expansion). *Uniformly in the regime of Theorem 1.2,*

$$\frac{A(N, k)}{B(N, k)} = e^{-2\rho} \left[1 + \frac{\rho - \rho^2}{k} + \frac{\rho^4/2 - 7\rho^3/3 + 3\rho^2 + \rho/3}{k^2} + \frac{-\rho^6/6 + 11\rho^5/6 - 41\rho^4/6 + 31\rho^3/3 - 2\rho^2/3 - 2\rho}{k^3} + O_I(k^{-4}) \right]. \quad (8.2)$$

All higher coefficients follow by the same substitution and subtraction.

Proof. Apply (1.10) to (N, k) and $(N, k + 1)$, use (8.1), and expand in $1/k$. The remainders stay uniform because ρ' remains in a slightly enlarged compact interval inside $(0, \infty)$. Collecting terms gives the displayed polynomials. $\square$

Corollary 8.2 (Conditional overshoot). *For every fixed integer $d \geq 1$, under a uniformly random permutation of N letters,*

$$\mathbb{P}(\text{LIS} \geq k + d \mid \text{LIS} \geq k) = \left(\frac{\rho}{k}\right)^d \left(1 + O_{I,d}(k^{-1})\right). \quad (8.3)$$

In particular, the conditional probability that $\text{LIS} = k$ is $1 - \rho/k + O_I(k^{-2})$.

Proof. The conditional probability is $T(N, k + d)/T(N, k)$. The ratio of its B factors is $(N - k)^d / ((k + 1)^d)^2$, which equals $(\rho/k)^d (1 + O_{I,d}(k^{-1}))$. The exponential factors and the relative correction series have ratio $1 + O_{I,d}(k^{-1})$. $\square$

Corollary 8.3 (Mean number of witnesses). *Let $Z_k(\pi)$ be the number of increasing subsequences of length k . Then*

$$\mathbb{E}[Z_k \mid \text{LIS} \geq k] = e^{2\rho} \left(1 - \frac{2\rho - \rho^2}{k} + O_I(k^{-2})\right). \quad (8.4)$$

Proof. Equation (5.8) gives $\mathbb{E}[Z_k \mid \text{LIS} \geq k] = B/T$. Invert (1.10). $\square$

This conclusion concerns the mean only. It does not imply a Poisson limit, concentration, or convergence of the entire witness-count distribution. Those require joint witness information not contained in the first mean.

9 OEIS diagonals and numerical checks

9.1 Five correction terms for A269021

Set $N = 2n$, $k = n$, so $\rho = 1$. The coefficient rule gives

$$(c_0(1), \dots, c_5(1)) = \left(1, 1, \frac{9}{2}, 7, -\frac{149}{8}, -\frac{3911}{40}\right). \quad (9.1)$$

Stirling's expansion gives

$$\frac{B(2n, n)}{16^n (n-1)!/\pi} = 1 - \frac{1}{4n} + \frac{1}{32n^2} + \frac{1}{128n^3} - \frac{5}{2048n^4} - \frac{23}{8192n^5} + O(n^{-6}). \quad (9.2)$$

One way to check it is to exponentiate $-1/(4n) + 1/(96n^3) - 1/(320n^5) + O(n^{-7})$.

Corollary 9.1 (A269021). For $a_n = T(2n, n)$,

$$a_n = \frac{16^n(n-1)!}{\pi e^2} \left[1 + \frac{3}{4n} + \frac{137}{32n^2} + \frac{757}{128n^3} - \frac{41429}{2048n^4} - \frac{3803959}{40960n^5} + O(n^{-6}) \right]. \quad (9.3)$$

The expansion continues to every fixed order by (7.4)–(7.7) and Stirling’s series.

Proof. Multiply (9.1), interpreted as its $1/n$ expansion from Theorem 1.2, by (9.2). $\square$

For the exact-length diagonal $b_n = A(2n, n)$, the same calculation using (8.2) gives

$$b_n = \frac{16^n(n-1)!}{\pi e^2} \left(1 - \frac{1}{4n} + \frac{49}{32n^2} + \frac{273}{128n^3} + O(n^{-4}) \right). \quad (9.4)$$

The different $1/n$ coefficients distinguish “at least n ” from “exactly n ”, despite their common leading equivalent. The latter leading equivalent is already proved in [7].

9.2 Independent exact comparisons

The supplied verifier uses the hook-length formula and integer partitions to compute A , T , and $C_1, \dots, C_4$ directly. A separate calculation uses truncated rational power series in the squared-minor formula (3.6). It checks genuinely nonzero $j = 2$ and $j = 3$ cases, not just empty supports.

n	$T(2n, n)$	$A(2n, n)$
0	1	1
1	2	1
2	23	13
3	588	381
4	24 553	17 557
5	1 438 112	1 100 902
6	108 469 917	87 116 283
7	9 996 042 284	8 312 317 976
8	1 086 997 811 325	927 716 186 325

The initial values agree with the corresponding OEIS entries [10, 11]; the small-index empty-permutation convention is imposed separately from the positive- k array.

The completed exact test run covers $0 \leq N \leq 26$, $1 \leq k \leq N + 1$: 378 cells. It verifies Plancherel normalization, the first-moment single sum, the residual identity, the sharp cutoff, and the first rectangle discrepancy for truncation ranks 1 through 4, whenever the relevant point lies in that range. There are 140 independent Bessel-minor comparisons. The A269021 formula matches 16 displayed OEIS terms and generates terms through $n = 60$. Symbolic computation produces the coefficient polynomials through order five and checks 15 finite-product specializations by a separate rational-series method.

9.3 Accuracy of the asymptotic expansion

Let $L_n = 16^n(n-1)!/(\pi e^2)$, and let $S_M(n)$ retain terms through n^{-M} in the square brackets of (9.3). The next table reports $|L_n S_M(n)/a_n - 1|$. The integer a_n is evaluated exactly; the rescaling uses 100 decimal digits of floating-point precision.

n	$M = 0$	$M = 1$	$M = 3$	$M = 5$
20	$4.65 \cdot 10^{-2}$	$1.08 \cdot 10^{-2}$	$1.44 \cdot 10^{-4}$	$4.38 \cdot 10^{-6}$
50	$1.65 \cdot 10^{-2}$	$1.73 \cdot 10^{-3}$	$3.46 \cdot 10^{-6}$	$1.62 \cdot 10^{-8}$
100	$7.87 \cdot 10^{-3}$	$4.30 \cdot 10^{-4}$	$2.10 \cdot 10^{-7}$	$2.40 \cdot 10^{-10}$
200	$3.84 \cdot 10^{-3}$	$1.07 \cdot 10^{-4}$	$1.29 \cdot 10^{-8}$	$3.64 \cdot 10^{-12}$
400	$1.90 \cdot 10^{-3}$	$2.68 \cdot 10^{-5}$	$7.98 \cdot 10^{-10}$	$5.60 \cdot 10^{-14}$

These floating-point comparisons are diagnostic, not interval-certified error bounds. The proof of the asymptotic statement is the uniform remainder argument, not agreement in the table. The result is an asymptotic expansion: adding terms need not improve an approximation at every small n or at arbitrarily high truncation order.

10 Further research questions and topics

The following questions are separate from the theorems proved above. They are ordered roughly by proximity to the present construction, not by a claim of difficulty or priority.

1. **Certify the specific one-third-sector recurrences.** Use the $C_1 - C_2$ representation to construct certificates for the operators guessed in [1]. The existence theorem does not itself identify those operators. A complete result would publish the certificates, initial region, and singular-index treatment.
2. **Determine recurrence complexity as the sector widens.** Find upper and lower bounds on minimal pure-shift orders, degrees, and holonomic ranks as functions of R . The fixed-dimensional multisum proves finiteness, but the naive determinant expansion has $(R!)^2$ outer terms at its largest moment. Can a structured representation avoid that growth?
3. **Compute higher moments without full determinant expansion.** Exploit Cauchy–Binet, symmetric-polynomial coordinates, or discrete orthogonality to obtain efficient exact formulas for C_2 , C_3 , and beyond. Any claimed speedup should distinguish arithmetic-operation counts from bit complexity and should be compared with independent tableau counts.
4. **Resolve the exponentially small sectors.** Find the first asymptotic equivalent of $C_1 - T$ in a region with $N > 2k + 2$, and then of the residual after R moments. The bound (6.2) does not determine the exponential action, prefactor, or all-order corrections. Near a rectangle boundary, a fixed excess number of boxes may offer a tractable starting regime.
5. **Extend uniformity beyond compact linear regimes.** Investigate $\rho \rightarrow 0$ with growing deficit, and $\rho \rightarrow \infty$ with $k = o(N)$. These are not covered by Theorem 1.2. The polynomial coefficients alone do not justify substitution of a growing ρ ; one needs new remainder estimates and, eventually, a different balance of moment contributions.

6. **Study the conditional witness distribution.** The mean number of length- k increasing subsequences tends to $e^{2\rho}$. Determine higher conditional moments, limiting laws, and structural descriptions of permutations in the rare tail event. Shifted-row factorial moments and witness-count factorial moments must be kept distinct.
7. **Classify rational-ray OEIS families.** Generate exact values and asymptotic expansions for $T(an + a_0, bn + b_0)$ and $A(an + a_0, bn + b_0)$, and identify which already have OEIS entries. The rational-ray theorem supplies a recurrence existence guarantee before guessing begins. Each proposed entry or formula should receive its own source and duplicate audit.
8. **Formalize the finite proof boundary.** A staged formalization could first verify the rectangle lemma, binomial residual, exact finite coefficient identity, and low-rank certificates. A full formalization must additionally connect Robinson–Schensted, the Bessel correlation theorem, and holonomic closure to the precise array semantics. Importing any of these as an axiom should be explicitly recorded rather than hidden by a finite numerical checker.

11 Conclusion and proof audit

The fixed-sector theorem follows from a short conceptual chain:

$$\text{shifted-row threshold} \implies \text{rectangle obstruction} \implies \text{finite inclusion–exclusion},$$

followed by the Bessel-kernel conversion of every fixed moment into a proper-hypergeometric multisum. The number of summation variables is fixed before N, k vary. This is the precise point that allows holonomic closure and establishes all fixed sectors.

The first moment has a much simpler single-sum expression. Its normalized summands converge to a signed exponential series, while factorial domination makes the expansion uniform to every algebraic order. The higher shifted-row contributions are either forbidden exactly or exponentially smaller than those orders. This gives a single coefficient calculus for the full positive-linear regime, including the two OEIS diagonals and their local probability consequences.

For audit purposes, the proof has four distinct layers. The external inputs are established named theorems, cited where used. The algebraic reductions are displayed explicitly, including the factorial arguments and support conditions needed by holonomic summation. The asymptotic claims include uniform remainder bounds. The executable tests provide independent finite checks, not a replacement for any infinite theorem. No specific high-order guessed recurrence, global holonomicity of the original array, or formal proof-assistant certification is claimed.

A Reproduction guide and package contents

The package is self-contained for its computations and does not need network access. It contains the article source and compiled PDF, `verify.py`, a requirements file, a source-audit note, and machine-readable results. Run

```
python verify.py --output results
```

The exact combinatorial checks need only Python’s standard library. The symbolic and high-precision parts use SymPy and mpmath. To run only the exact checks, use

```
python verify.py --core-only --output results
```

The source can be compiled with

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The principal output files are

File under results/	Contents
small_counts.csv	Exact-length counts, tails, and four moments for small N, k .
bessel_checks.csv	Independent rational-series evaluations of fixed moments.
a269021.csv, a267433.csv	Exact tail and exact-length diagonal values for $0 \leq n \leq 60$.
symbolic_coefficients.json	The five computed polynomial corrections and diagonal specializations.
asymptotic_checks.csv	Signed relative errors of four asymptotic truncations.
uniform_checks.csv	Off-diagonal tests at $\rho = 1/2, 1, 2$.
verification_summary.json	Test scope and pass status.

The Bessel verifier factors $J_d(2\sqrt{t}) = t^{d/2}F_d(t)$, where $F_d(t) = \sum_s (-1)^s t^s / (s!(d+s)!)$. For each increasing pair u, v with budget (3.7), it forms $\det F$, squares it, and convolves with e^t . All of this is done with exact rational coefficients. The independent tableau computation uses no Bessel functions. This separation detects normalization, shift, missing factorial, and determinant-sign errors that would be difficult to spot from a single formula alone.

[write] Added 3 October 2026, batch 85. This appendix describes the delivered package. In ProveIt the verifier is `code/verify.py`, the files listed under `results/` are shipped in `data/`, beside `data/requirements.txt`, and `build.sh` is `code/build.sh` (it runs three `pdflatex` passes, not the two shown above); the source-audit note is `SOURCE_AUDIT.md`. The delivered PDF is not shipped; the committed `article.pdf` is a rebuild including these notes. Run from `code/` without `--output`, the verifier would write a new `code/results/` directory; the report's README gives a rerun on a scratch copy that never writes into the report. The sixteen terms each of A269021 and A267433 embedded in the verifier are OEIS data, under CC BY-SA 4.0. At intake (3 October 2026) the full run passed on such a copy in 71 s (the delivered summary records 4.6 s on the producer's machine); every regenerated data file matched the shipped one, the two text-mode outputs up to line endings, and the summary differed only in its elapsed time.

B Additional coefficient polynomials

The fourth and fifth uniform tail coefficients, generated by the all-orders rule and checked by the supplied symbolic code, are

$$c_4(\rho) = \frac{\rho}{360} (15\rho^7 - 360\rho^6 + 3440\rho^5 - 16716\rho^4 + 38960\rho^3 - 19080\rho^2 - 15100\rho + 2136), \quad (\text{B.1})$$

$$c_5(\rho) = -\frac{\rho}{360} (3\rho^9 - 110\rho^8 + 1670\rho^7 - 13796\rho^6 + 66352\rho^5 - 170464\rho^4 + 145380\rho^3 + 67016\rho^2 - 54612\rho - 6240). \quad (\text{B.2})$$

These expressions concern the normalization $B(N, k)e^{-2\rho}$. They should not be substituted directly into the A269021 normalization $16^n(n-1)!/(\pi e^2)$ without also applying the central-binomial correction (9.2).

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Web and repository source audit: October 3, 2026. OEIS entries are living documents; the audit records the versions inspected, not an assertion about all subsequent updates.